

Mahesh Bhosale

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Homepage

LinkedIn

Github

Research Interest

Image/Video synthesis via Diffusion Models, VQA and Fairness issues of Multi-Modal Large Language Models (MLLMs).

Education

State University of New York at Buffalo, Buffalo, NY, USA

Sept 2023 – May 2027 (Expected)

PhD in Computer Science

- Advisor: [Dr. David Doermann](#)
- Google CS Research Mentorship Scholar 2023.

State University of New York at Buffalo, Buffalo, NY, USA

Sept 2021 – May 2023

MS by Research in Computer Science

- Advisor: [Dr. David Doermann](#)
- Projects: Person Re-identification in Videos.

Walchand College of Engineering, Shivaji University, Sangli, India

June 2013 – May 2017

BS in Information Technology

Publications

[Google Scholar](#)

- CVPR 2026** [Mahesh Bhosale](#), Abdul Wasi*, Shantam Shrivastava*, Shifa Latif, Tianyu Luan, Mingchen Gao, David Doermann, Xuan Gong, FairLLaVA: Fairness-Aware Parameter-Efficient Fine-Tuning for Large Vision-Language Assistants, in IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026. [\[Paper\]](#) [\[Code\]](#)
- MIDL 2026** Shantam Shrivastava, [Mahesh Bhosale](#), David Doermann, Mingchen Gao, Category-wise Structured Radiology Report Generation with Contrastive Decoding, Medical Imaging with Deep Learning, 2026. (**Oral** $\leq 8\%$ of submitted papers) [\[Paper\]](#)
- ACL 2026** [Mahesh Bhosale*](#), Abdul Wasi*, Vishvesh Trivedi*, Pengyu Yan, Akhil V S S Gorugantu, David Doermann, CRAFT: Critic-Refined Adaptive Key-Frame Targeting for Multimodal Video Question Answering, in Association for Computational Linguistics, Multimodal Augmented Generation via Multimodal Retrieval Workshop, 2026. [\[Paper\]](#) [\[Code\]](#)
- ACL 2026** Pengyu Yan*, Akhil V S S Gorugantu*, [Mahesh Bhosale](#), Abdul Wasi, Vishvesh Trivedi, David Doermann, TRACE: Evidence Grounding-Guided Multi-Video Event Understanding and Claim Generation, in Association for Computational Linguistics, Multimodal Augmented Generation via Multimodal Retrieval Workshop, 2026. [\[Paper\]](#) [\[Code\]](#)
- Under Review** [Mahesh Bhosale*](#), Naresh Kumar Devulapalli*, Abdul Wasi, Chau Pham, Vishnu Lokhande, David Doermann, Variance-Guided Score Regularization for Hallucination Mitigation in diffusion models, 2026.
- Under Review** Abdul Wasi, Akhil V S S Gorugantu, [Mahesh Bhosale](#), Sauradip Nag, Anjan Dutta, Junsong Yuan, David Doermann, RADI3N: Training-Free Geometric Image Editing via Parametric 3D Control, 2026.
- ICCV 2025** [Mahesh Bhosale](#), Abdul Wasi, Yuanhao Zhai, Yunjie Tian, Samuel Border, Nan Xi, Pinaki Sarder, Junsong Yuan, David Doermann, Xuan Gong, PathDiff: Histopathology Image Synthesis with Unpaired Text and Mask Conditions, in International Conference on Computer Vision, 2025. [\[Paper\]](#) [\[Code\]](#)

- NeurIPS 2025** Chau Pham, Quan Dao, **Mahesh Bhosale**, Yunjie Tian, Dimitris N. Metaxas, David Doermann, AutoEdit: Automatic Hyperparameter Tuning for Image Editing, in Neural Conference on Neural Information Processing Systems, 2025. [\[Paper\]](#) [\[Code\]](#)
- ICDAR 2024** Pengyu Yan, **Mahesh Bhosale**, Jay Lal, Bikhyat Adhikari, and David Doermann, ChartReformer: Natural Language-Driven Chart Image Editing, in International Conference on Document Analysis and Recognition, 2024. [\[Paper\]](#) [\[Code\]](#)
- ICDAR 2023** Jay Lal, **Mahesh Bhosale***, Aditya Mitkari*, and David Doermann, LineFormer: Rethinking Line Chart Data Extraction as Instance Segmentation, in International Conference on Document Analysis and Recognition, 2023. [\[Paper\]](#) [\[Code\]](#)
- arXiv 2023** **Mahesh Bhosale***, Abhishek Kumar*, and David Doermann, Player Re-Identification Using Body Part Appearances, arXiv 2023. [\[Paper\]](#) [\[Code\]](#)

On-going Research Projects

- **Id-Diff**: Exploring relationships between intrinsic dimension of data and generative performance of latent diffusion models. (In collaboration with **Carez AI and University of Montreal**).
- **AI agents for Pathology**: Exploring agentic AI for computational pathology that predicts patient survival while remaining clinically interpretable. (In collaboration with **NYU Langone Health**).
- **Few Step Distillation of Flow Models**: Trying to tackle the issues of training GAN in distillation.(In collaboration with **Meta**.)
- **Multi-Video Understanding**: Tackling efficiency and grounding of article generation from multiple videos given query. (In collaboration with **Johns Hopkins University**.)

Experience

Visiting Research Scholar

June 2026 – August 2026

[Johns Hopkins University, Baltimore, MD, USA](#)

- Research Topics: Multi-video understanding.

Research Assistant and Lab Manager

May 2022 – Present

[Institute of Artificial Intelligence and Data Science, University at Buffalo, Buffalo, NY, USA](#)

- Research Topics: Multi-modal generative AI

Software Engineer

Aug 2017 – Aug 2021

[Veritas Technologies LLC, Pune, India](#)

- Advisor: [Anindya Banerjee](#). Developed a novel scheduling algorithm on top of LSTM storage and compute resource prediction. Reduced execution time of resource intensive tasks by 56%. Bench-marked other Deep learning based sequence modeling solutions. Bestowed Certificate of Merit Veritas Illuminate 2020.

Software Engineer Intern

Jan 2017 – June 2017

[Veritas Technologies LLC, Pune, India](#)

- Reduced execution time of variants of the 'ls' command by 20% using directory inode read ahead algorithm.

Invited Talks

Introduction to Image Generative Models [\[Slides\]](#)

University at Buffalo, Oct 2025

Automated analysis of Sporting Event Videos Talk-1 [\[Slides\]](#)

University at Buffalo, Mar 2022

Automated analysis of Sporting Event Videos Talk-2 [\[Slides\]](#)

University at Buffalo, Apr 2022

Soccer Player Re-identification [\[Slides\]](#)

University at Buffalo, May 2022

Professional Services

Teaching Assistant

CSE 573 Computer Vision and Image Processing, Fall 2025
CSE 574 Introduction to Machine Learning, Spring 2024, Fall 2024, Spring 2025
CSE 702 Automated Analysis of Sporting Event Videos, Summer 2022
CSE 521 Introduction to Operating Systems, Spring 2022, Fall 2022, Fall 2023

Conference Reviewer

IEEE International Conference on Computer Vision (ICCV) 2025
Neural Information Processing Systems (NeurIPS) 2026
International Conference on Learning Representations (ICLR) 2026
Conference on Computer Vision and Pattern Recognition (CVPR) 2026

Honors and Awards

Research Mentorship Scholar , Google, USA (remote)	2023
Certificate of Merit , Illuminate Annual Technical Conference, Veritas Technologies LLC, Pune, India	2020
Silver Medal , World Code Sprint, Hacker-rank	2016
Honorable Mention , ACM-ICPC Amritapuri Regional, Amrita University, Coimbatore, India	2015
Runner Up (out of 75 teams), Best Project Competition, Department of IT, WCE, Sangli, India	2015
Runner Up (600 participants), Coding Competition, MindSpark, College of Engineering, Pune, India	2014
DST Inspire Science Camp (Top 1%), Government of India and IIIT, Pune, India	2014